Identifying Cancer-Related Fatigue Trajectories in Breast Cancer Survivors

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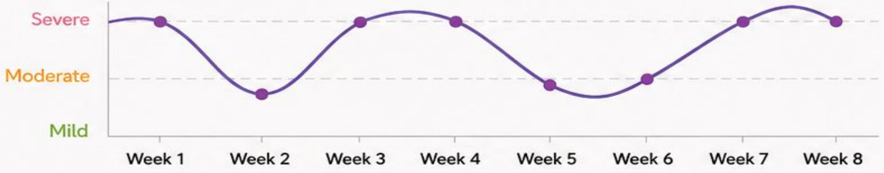
Problem statement: Cancer-Related Fatigue in Cancer Survivors

>40%

of cancer survivors experience **cancer-related fatigue**

Fatigue varies over time



Cancer Survivor → Weekly HA-CRF Questionnaire → Continuous Fitbit Data (Activity, Sleep, Heart Rate) → Machine Learning Model → Fatigue Trajectory Types

What is CRF?

- Affects >40% of cancer survivors
- Persistent exhaustion not relieved by rest
- Impacts daily functioning & quality of life

Why is it difficult to study?

- Varies substantially between individuals
- Varies within the same individual over time
- Heterogeneous in severity & symptoms
- Most studies measure months apart, missing weekly/daily fluctuations

This Project Aims To:

- Identify distinct fatigue trajectory types using weekly HA-CRF questionnaires over 8 weeks
- Examine whether continuous Fitbit data differ between fatigue types
- Predict fatigue types from wearable data using a suitable machine learning model

Preliminary overview of related works

This project builds on three earlier studies. Together, they show that CRF is common, varies between individuals, and can be measured in detail.

1 Bower (2014) – The clinical picture

- CRF affects more than 40% of cancer survivors
- It is persistent, disabling, and different from normal tiredness
- Linked to inflammation, sleep problems, and other biological changes

2 Thong et al. (2017) – Different types of fatigue

- Found three distinct CRF groups in a large patient registry
- Each group had a different pattern of severity and symptoms

How is it relevant to our project?

- Their questionnaire is different from ours
- We may not find the same three types
- We will identify fatigue types from our own data

3 Wijlens et al. (2024) – Measuring CRF in detail

HA-CRF

Wijlens and colleagues (2024) created the Holistic Assessment of CRF (HA-CRF) questionnaire. It measures daily activity patterns, social health, coping strategies, and other factors that keep fatigue going.

- Daily activity patterns
- Social health
- Coping strategies
- Factors maintaining fatigue

Our dataset

- 42 female breast cancer survivors
- Weekly HA-CRF for 8 weeks → 378 questionnaire data points
- Fitbit Inspire 3 data (continuous activity, sleep, heart rate)

How we build on this work

Combine weekly questionnaires with continuous Fitbit data → Account for repeated measures (9 time points per person) → Use machine learning to predict fatigue types from Fitbit data

The Data Challenge

The two data sources operate at different speeds and cannot be directly combined.

- **Questionnaire data:** One fatigue score per patient per week via the HA-CRF questionnaire
- **Fitbit data:** Continuous physiological measurements 24 hours a day: heart rate, step counts, and sleep stages
- **The problem:** Merging the datasets. High-frequency sensor data cannot be directly mapped to a weekly questionnaire score without proper alignment
- **Our solution:** For each weekly questionnaire, the preceding 7 days of Fitbit data are summarized into weekly features: step trends, heart rate variance, and sleep efficiency
- **The result:** One clean observation per questionnaire entry, ready for analysis

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Discovering Fatigue Trajectories

- **The Problem with Averages:** Averaging fatigue scores across all 42 patients hides the distinct individual groups
- **The Analytical Approach:** Group-Based Trajectory Modeling
- **How it Works:** Scans the 8 weeks of survey data and clusters patients who follow similar fatigue pathways over time
- **The Output:** Transforms survey scores into clear labeled groups (e.g. "Mild and Stable" vs "Severe and Persistent") which become the **targets** for our predictive model
- **Why this method?** More flexible alternatives (Growth Mixture Modeling) requires a larger sample and 42 patients is too small

Machine Learning Model Selection: Mixed Effects Random Forest

Standard models are limited. They treat each weekly measurement as a different person so they memorize patients, not fatigue

Mixed Effects Random Forest solves this:

- Random Forest layer: learns fatigue patterns true for everyone (declining steps, poor sleep)
- Mixed Effects layer: accounts for each patient's personal baseline (one patient just naturally has a higher heart rate, so that is not fatigue)

Validation: We will use Group K-Fold Cross-Validation so the model is tested only on patients it has never seen before, ensuring results generalise beyond our 42 participants

Time Planning (May 20 – June 24)

Sprint 1 (May 20 to May 26): Target Definition

- Clean the weekly questionnaire data and handle missing values
- Run Group-Based Trajectory Modeling to discover fatigue groups
- Assign a fatigue group label to each patient (e.g. "Mild-Stable" or "Severe-Persistent")

Sprint 2 (May 27 to June 2): Feature Engineering

- Clean the continuous Fitbit data and remove gaps and noise
- Build the 7-day rolling window to align Fitbit data with each weekly questionnaire
- Extract weekly summaries: step trends, sleep efficiency, heart rate variance
- Merge the engineered Fitbit features with the group labels into one final dataset ready for training

Sprint 3 (June 3 to June 9): Model Training

- Load the final dataset for training
- Train the Mixed Effects Random Forest in Python
- Record initial performance results

Sprint 4 (June 10 to June 16): Validation and Tuning

- Run Group K-Fold Cross-Validation on unseen patients
- Tune model parameters to improve accuracy
- Assess and evaluate the results

Sprint 5 (June 17 to June 23): Presentation and Paper Prep

- Finalize slides and speaking notes
- Rehearse presentation

Presentation Day: June 24

ANY QUESTIONS?